Bases, Dimension, and Rank

Linear Algebra · Lecture 4

Last time (Lecture 3): subspaces, spans, linear independence and bases. Here $A = [a_1 \ \cdots \ a_n] \in \mathbb{R}^{m\times n}$ has the columns a_j; e_j is column j of I; row_i, col_j are row i and column j.

From Lecture 1, Lecture 2 and Lecture 3 (restated).

- Lecture 1, Theorems 3(a), 8, 4(a)–(c), Definition 7: $(AB)C = A(BC)$, $A(B+C) = AB + AC$, $c(AB) = (cA)B = A(cB)$, $I_m A = A = AI_n$, $A0 = 0$; $(A')_{ij} = a_{ji}$, $(A')' = A$, $(A+B)' = A' + B'$, $(cA)' = cA'$, $(AB)' = B'A'$; $x'y = y'x = \sum_i x_i y_i$, $x'x = x_1^2 + \cdots + x_n^2$; $\text{col}_j(AB) = A\,\text{col}_j(B)$; $Ax = x_1 a_1 + \cdots + x_n a_n$, $Ae_j = a_j$; $\text{row}_i(AB) = \text{row}_i(A)B = a_{i1}\,\text{row}_1(B) + \cdots + a_{in}\,\text{row}_n(B)$.
- Lecture 2, Definitions 2, 4, 5: row operations $R_r \leftarrow cR_r$ $(c \neq 0)$, $R_r \leftarrow R_r + cR_s$, $R_r \leftrightarrow R_s$ $(r \neq s)$. R is in **RREF** if (a) the first nonzero entry of each nonzero row is 1, its **pivot**; (b) a column with a pivot has all other entries 0; (c) zero rows lie below nonzero rows; (d) the pivots of the nonzero rows $1, \ldots, r$ lie in columns $k_1 < \cdots < k_r$, the **pivot columns**; so $\text{col}_{k_i}(R) = e_i$. **Echelon form**: (a), (c), (d). Gauss–Jordan: forward elimination gives an echelon form; then backward elimination, for each pivot from the last to the first, adds multiples of its row to the rows above it to make the entries above the pivot 0. An **RREF** (**echelon form**) **of** A: a matrix in RREF (echelon form) reached from A by row operations.
- Lecture 2, Theorem 3(b), Proposition 7: if row operations take $[A \mid b]$ to $[R \mid z]$, then $Ax = b$ and $Rx = z$ have the same solutions. Let R be in RREF, with nonzero rows $1, \ldots, r$, pivot columns $k_1 < \cdots < k_r$, **pivot unknowns** x_{k_i}, the others **free unknowns**. (a) $Ax = b$ is consistent $\iff$ $z_i = 0$ for all $i > r$. (b) If so, the solutions: free unknowns arbitrary, $x_{k_i} = z_i - \sum_{j \text{ free}} R_{ij} x_j$ $(i \leqslant r)$.
- Lecture 2, Definition 10, Theorems 14(c), 15, 22(a): $P \in \mathbb{R}^{m\times m}$ is invertible if $PQ = QP = I_m$ for some $Q = P^{-1}$. If row operations take A to B, then $B = PA$, P invertible. $A \in \mathbb{R}^{n\times n}$ is invertible $\iff$ $Ax = 0$ only for $x = 0$. If $Ax_p = b$, the solutions of $Ax = b$ are the $x_p + z$ with $Az = 0$.
- Lecture 3, Definition 3, Theorem 4, Example 5(b): a **subspace** $W \subseteq \mathbb{R}^n$ (or $\mathbb{R}^{1\times n}$) is a vector space with the same operations: $0, u+v, cu \in W$ $(u, v \in W,\ c \in \mathbb{R})$; $\{x : Ax = 0\}$ is one.
- Lecture 3, Definition 6, Theorems 7, 9(c): $\text{span}\{v_1, \ldots, v_k\} = \{\sum_i c_i v_i : c_i \in \mathbb{R}\}$ ($\text{span}\,\varnothing = \{0\}$) is the smallest subspace containing $v_1, \ldots, v_k$; adjoining vectors of $\text{span}\{a_1, \ldots, a_n\}$ keeps the span.
- Lecture 3, Definitions 10, 14, Propositions 11, 12(a),(b): $v_1, \ldots, v_k$ are **linearly independent** if $c_1 v_1 + \cdots + c_k v_k = 0$ only for $c_1 = \cdots = c_k = 0$ (the empty list is independent), otherwise **dependent**. For $v_i \in \mathbb{R}^m$, $V = [v_1 \ \cdots \ v_k]$: independent $\iff$ $Vc = 0$ only for $c = 0$ $\iff$ every column of an RREF of V is a pivot column; more than m vectors of $\mathbb{R}^m$ are dependent. A list containing 0 is dependent; sublists of independent lists are independent. A **basis** of W: independent vectors of W that span W; $e_1, \ldots, e_n$ of $\mathbb{R}^n$ (standard basis); $\varnothing$ of $\{0\}$.

1. Bases and dimension of subspaces of $\mathbb{R}^n$

Theorem 1. Let R be an RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows. The following are equivalent: (i) the columns $a_1, \ldots, a_n$ of A are linearly independent; (ii) $Ax = 0$ only for $x = 0$; (iii) $r = n$, that is, every column of R is a pivot column. In particular, if $n > m$, then any n vectors of $\mathbb{R}^m$ are linearly dependent.

Proof.

Lecture 3, Proposition 11, with $V = A$; Lecture 2, Definition 4(a),(d):

(i) $\iff$ (ii) $\iff$ every column of R is a pivot column; $\quad n > m \implies$ dependent

each nonzero row of R has one pivot, in different columns $\implies$ r pivot columns

every column of R is a pivot column $\iff$ $r = n$

□

Theorem 2. Let R be an RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows, pivot columns $k_1 < \cdots < k_r$ and entries R_{ij}.

(a) For $x \in \mathbb{R}^n$: $x_1a_1 + \cdots + x_na_n = 0 \iff x_1\,\mathrm{col}_1(R) + \cdots + x_n\,\mathrm{col}_n(R) = 0$.

(b) If $r \geqslant 1$, the **pivot columns** $a_{k_1}, \ldots, a_{k_r}$ of A form a basis of $\mathrm{span}\{a_1, \ldots, a_n\}$, and

$$a_j = R_{1j}a_{k_1} + R_{2j}a_{k_2} + \cdots + R_{rj}a_{k_r} \qquad (j = 1, \ldots, n)$$

If $r = 0$, then $A = 0$ and $\mathrm{span}\{a_1, \ldots, a_n\} = \{0\}$.

In particular, every finite list that spans a subspace W contains a basis of W.

Proof.

(a) Lecture 1, Theorem 4(b); Lecture 2, Theorem 3(b) for $[A \mid 0] \to [R \mid 0]$:

$$x_1a_1 + \cdots + x_na_n = Ax, \quad x_1\,\mathrm{col}_1(R) + \cdots + x_n\,\mathrm{col}_n(R) = Rx; \qquad Ax = 0 \iff Rx = 0$$

(b) $r = 0$: $R = 0$, and $R = PA$ with P invertible (Lecture 2, Theorem 14(c)), so $A = P^{-1}R = 0$. Let $r \geqslant 1$.

Relations: $\mathrm{col}_{k_i}(R) = e_i$, and rows $r+1, \ldots, m$ of R are 0:

$\mathrm{col}_j(R) = \sum_{i=1}^r R_{ij}e_i = \sum_{i=1}^r R_{ij}\,\mathrm{col}_{k_i}(R)$ (every j)

$j \notin \{k_1, \ldots, k_r\}$: $x_j = 1$, $x_{k_i} = -R_{ij}$, other $x_l = 0 \implies Rx = \mathrm{col}_j(R) - \sum_i R_{ij}\,\mathrm{col}_{k_i}(R) = 0$

$\implies Ax = a_j - \sum_i R_{ij}a_{k_i} = 0$ ((a))

$j = k_l$: $R_{ik_l} = 1$ $(i = l)$, 0 $(i \neq l) \implies$ the formula reads $a_{k_l} = a_{k_l}$

Basis: $U = \mathrm{span}\{a_{k_1}, \ldots, a_{k_r}\}$:

$a_1, \ldots, a_n \in U \implies \mathrm{span}\{a_1, \ldots, a_n\} \subseteq U \subseteq \mathrm{span}\{a_1, \ldots, a_n\}$ (Lecture 3, Theorem 7)

$c_1a_{k_1} + \cdots + c_ra_{k_r} = 0 \implies c_1e_1 + \cdots + c_re_r = 0 \implies c_1 = \cdots = c_r = 0$ ((a))

Last sentence: write the spanning vectors as the columns of a matrix and apply (b); if $r = 0$, then $W = \{0\}$, and the empty list is a basis of W. □

Pivot columns of A. Let R be an RREF of A, with pivot columns $k_1 < \cdots < k_r$.

- By Theorem 2(a) (take $x_j = -1$, $x_l = 0$ for $l > j$), a_j is a linear combination of $a_1, \ldots, a_{j-1} \iff$ $\mathrm{col}_j(R)$ is a linear combination of $\mathrm{col}_1(R), \ldots, \mathrm{col}_{j-1}(R)$.
- In R this happens exactly when j is not a pivot column: $\mathrm{col}_{k_i}(R) = e_i$, while the columns before k_i are 0 in rows $i, \ldots, m$ (Lecture 2, Definition 4(a),(c),(d)); and a column $j \notin \{k_1, \ldots, k_r\}$ is $\sum_{k_i<j} R_{ij}\,\mathrm{col}_{k_i}(R)$ (proof of Theorem 2(b), first line; row i of R is 0 before column k_i).
- So the pivot columns of A are the columns that are not combinations of the earlier ones: they, and their number r, do not depend on the row operations used.

Remark 3. Let H be an echelon form of A, with pivots in columns $k_1 < \cdots < k_r$. Then the pivot columns of A are $a_{k_1}, \ldots, a_{k_r}$, and r is the number of nonzero rows of H. So for Theorem 1 one may stop at an echelon form: (iii) says that every column of H contains a pivot. The same holds for a matrix reached from A that satisfies (c), (d) of the definition of RREF with its first nonzero entries not yet scaled to 1.

Proof.

- Backward elimination adds multiples of a row i, whose pivot is in column k_i, to rows $l < i$, whose pivot is in a column $k_l < k_i$. Row i is 0 before column k_i, so row l changes only in columns $\geqslant k_i$ and keeps its pivot in column k_l. Row i is also 0 in the pivot columns $k_{i+1}, \ldots, k_r$ already cleared, so they stay $e_{i+1}, \ldots, e_r$; the zero rows of H do not change. So the result is in RREF (Lecture 2, Definition 5).
- So backward elimination takes H to an RREF of A with the same pivot columns $k_1, \ldots, k_r$ as H; by the paragraph above, these are the pivot columns of A.

- Scaling a row by $c \neq 0$ does not move its first nonzero entry. □

Theorem 4. Every subspace W of $\mathbb{R}^n$ has a basis, and any two bases of W have the same number of elements, the **dimension** $\dim W$ of W. $\dim \mathbb{R}^n = n$ (standard basis), $\dim\{0\} = 0$ (empty list), and $\dim W = 0$ only for $W = \{0\}$. If W is spanned by m vectors, then any $k > m$ vectors of W are linearly dependent.

Proof.

$W = \operatorname{span}\{w_1, \dots, w_m\}$, $v_1, \dots, v_k \in W$, $k > m$ (if $m = 0$: $W = \{0\}$, and $k \geqslant 1$ vectors of $\{0\}$ are dependent, Lecture 3, Proposition 12(a)); $m \geqslant 1$:

$$v_j = a_{1j}w_1 + \cdots + a_{mj}w_m \ (j = 1, \dots, k), \quad A = (a_{ij}) \in \mathbb{R}^{m \times k}$$
$$k > m \implies Ax = 0 \text{ for some } x \neq 0 \qquad \text{(Theorem 1)}$$
$$x_1v_1 + \cdots + x_kv_k = \textstyle\sum_{j=1}^{k} x_j \sum_{i=1}^{m} a_{ij}w_i = \sum_{i=1}^{m} (Ax)_i w_i = 0$$
$$\text{bases } v_1, \dots, v_k \text{ and } w_1, \dots, w_m \text{ of } W \implies k \leqslant m \text{ and } m \leqslant k$$

- Existence: start with the empty list and, as long as the vectors chosen do not span W, adjoin a vector $v \in W$ outside their span. In a relation $c_1v_1 + \cdots + c_kv_k + cv = 0$, $c \neq 0$ would put v in the span of $v_1, \dots, v_k$; so $c = 0$, and then every $c_i = 0$: the list stays independent.
- More than n vectors of $\mathbb{R}^n$ are dependent (Theorem 1), so this stops after at most n steps, at a basis of W.
- A basis with no vectors is the empty list, which spans $\{0\}$ (Lecture 3, Definition 6). □

Theorem 5. Let W be a subspace of $\mathbb{R}^n$ with $\dim W = d$.

(a) If U is a subspace of W, then $\dim U \leqslant d$, and $\dim U = d$ only for $U = W$. In particular $d \leqslant n$, and $d = n$ only for $W = \mathbb{R}^n$.

(b) Every linearly independent list of vectors of W can be extended to a basis of W.

(c) More than d vectors of W are linearly dependent; fewer than d vectors do not span W.

(d) d vectors of W are linearly independent $\iff$ they span W $\iff$ they form a basis of W.

Proof. $d = 0$: $W = \{0\}$ (Theorem 4).

- Every subspace U of W is $\{0\} = W$; a nonempty list of vectors of W contains 0, so it is dependent (Lecture 3, Proposition 12(a)); the empty list, the only independent one, is a basis of W (Lecture 3, Definition 14). So (a)–(c) hold.
- (d): the only list of 0 vectors is the empty list; it is independent, spans $\{0\} = W$ and is a basis of W.

Let $d \geqslant 1$, and let $w_1, \dots, w_d$ be a basis of W (Theorem 4). We prove (c) first.

(c) $w_1, \dots, w_d$ span W; Theorem 4, last sentence:

$$k > d \text{ vectors of } W \text{ are dependent} \qquad (m = d)$$
$$k < d \text{ vectors spanning } W \implies w_1, \dots, w_d \ (d > k) \text{ dependent } \text{✘} \qquad (m = k)$$

(b)

- Let $v_1, \dots, v_k \in W$ be independent. If they span W, they form a basis of W.
- Otherwise take $v \in W$ outside $\operatorname{span}\{v_1, \dots, v_k\}$. Then $v_1, \dots, v_k, v$ are independent: in $c_1v_1 + \cdots + c_kv_k + cv = 0$, $c \neq 0$ would put v in the span, so $c = 0$, and then every $c_i = 0$.

- Repeat. Each list obtained is independent in W, so it has at most d vectors by (c); the process stops, at a basis of W that contains $v_1, \dots, v_k$.

(a) A basis of U (Theorem 4) consists of $\dim U$ independent vectors of W:

$\dim U \leqslant d$ ((c))
$U \neq W,\ v \in W \setminus U \implies$ a basis of U, and v: $\dim U + 1$ independent vectors of W (as in (b))
$\implies \dim U + 1 \leqslant d$; so $\dim U = d$ only for $U = W$ ((c))
$W \subseteq \mathbb{R}^n,\ \dim \mathbb{R}^n = n \implies d \leqslant n,\ d = n$ only for $W = \mathbb{R}^n$ (Theorem 4)

(d) A basis of W is independent and spans W (Lecture 3, Definition 14). Let $v_1, \dots, v_d \in W$.

- If they are independent and $w \in W$: by (c), $x_0 w + x_1 v_1 + \cdots + x_d v_d = 0$ with not all $x_i = 0$; $x_0 \neq 0$, as $v_1, \dots, v_d$ are independent; so $w = -(x_1/x_0)v_1 - \cdots - (x_d/x_0)v_d$. They span W: a basis.
- If they span W: by Theorem 2, some of them form a basis of W, which has d vectors (Theorem 4); so it is all of $v_1, \dots, v_d$. □

Transposition carries a subspace of $\mathbb{R}^{1\times n}$ onto a subspace of $\mathbb{R}^n$ and preserves linear combinations, hence independence, spanning and bases; so Theorems 4 and 5 also hold for subspaces of $\mathbb{R}^{1\times n}$, such as the row space $\mathcal{R}(A)$ below.

Example 6. Let $W = \operatorname{span}\{u_1, \dots, u_5\} \subseteq \mathbb{R}^4$, where $u_1 = (1,2,1,1)'$, $u_2 = (1,3,2,1)'$, $u_3 = (1,1,0,1)'$, $u_4 = (1,2,2,3)'$, $u_5 = (1,3,1,-1)'$. (a) Are $u_1, \dots, u_5$ linearly independent? (b) Find a basis of W among $u_1, \dots, u_5$, and $\dim W$. (c) Express the remaining vectors in this basis.

Solution.

Row-reduce $A = [u_1 \ \cdots \ u_5]$:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 & 1 \\ 1 & 1 & 1 & 3 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 2 & -2 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 & -2 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_3} \begin{pmatrix} 1 & 1 & 1 & 0 & 2 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = R$$

$r = 3$, pivot columns $k_1 = 1$, $k_2 = 2$, $k_3 = 4$:

(a) No: $r = 3 < 5$; indeed any five vectors of $\mathbb{R}^4$ are dependent (Theorem 1)
(b) u_1, u_2, u_4 form a basis of W, $\dim W = 3$; $3 < 4 \implies W \neq \mathbb{R}^4$ (Theorems 2(b), 5(a))
(c) columns 3, 5 of R: $u_3 = 2u_1 - u_2$, $\quad u_5 = u_1 + u_2 - u_4$ (Theorem 2(b))

Check: $2(1,2,1,1)' - (1,3,2,1)' = (1,1,0,1)' = u_3$ and $(1,2,1,1)' + (1,3,2,1)' - (1,2,2,3)' = (1,3,1,-1)' = u_5$.

2. The four subspaces of a matrix

Definition 7. For $A \in \mathbb{R}^{m\times n}$, the **column space**, **row space**, **null space** and **left null space** of A are

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \qquad \mathcal{R}(A) = \{y'A : y \in \mathbb{R}^m\} \subseteq \mathbb{R}^{1\times n}$$
$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \qquad \mathcal{N}(A') = \{y \in \mathbb{R}^m : A'y = 0\}$$

The columns of A span $\mathcal{C}(A)$, the rows of A span $\mathcal{R}(A)$, and $y \in \mathcal{N}(A') \iff y'A = 0$ (transpose). All four are subspaces (Lecture 3, Theorem 7, Example 5(b)).

Let R be an RREF of A, with r nonzero rows, pivot columns $k_1 < \cdots < k_r$ and **free columns** $f_1 < \cdots < f_{n-r}$ (the free unknowns of $Ax = 0$ are $x_{f_1}, \ldots, x_{f_{n-r}}$). For $j = 1, \ldots, n-r$, the **special solution** $s_j \in \mathbb{R}^n$ is the solution of $Ax = 0$ with $x_{f_j} = 1$ and $x_{f_l} = 0$ $(l \neq j)$. By Lecture 2, Proposition 7(b), its entries are

$$(s_j)_{f_l} = \delta_{jl} \ (1 \leqslant l \leqslant n-r), \qquad (s_j)_{k_i} = -R_{if_j} \ (1 \leqslant i \leqslant r)$$

where $\delta_{jl} = 1$ if $j = l$ and 0 otherwise.

Theorem 8. Let $A \in \mathbb{R}^{m\times n}$, and let row operations take $[A \mid I_m]$ to $[R \mid P]$ with R in RREF (an RREF of A), with nonzero rows $\rho_1', \ldots, \rho_r'$ $(\rho_i \in \mathbb{R}^n)$, pivot columns $k_1 < \cdots < k_r$ and free columns $f_1 < \cdots < f_{n-r}$. Then $P \in \mathbb{R}^{m\times m}$ is invertible, $PA = R$, and:

(a) if $r \geqslant 1$, the pivot columns $a_{k_1}, \ldots, a_{k_r}$ of A form a basis of $\mathcal{C}(A)$;

(b) $\mathcal{R}(R) = \mathcal{R}(A)$, and if $r \geqslant 1$, the nonzero rows $\rho_1', \ldots, \rho_r'$ of R form a basis of $\mathcal{R}(A)$;

(c) the special solutions $s_1, \ldots, s_{n-r}$ form a basis of $\mathcal{N}(A)$ (if $r = n$: $\mathcal{N}(A) = \{0\}$, empty basis);

(d) $\mathrm{row}_{r+1}(P)', \ldots, \mathrm{row}_m(P)'$ form a basis of $\mathcal{N}(A')$ (if $r = m$: $\mathcal{N}(A') = \{0\}$, empty basis).

Hence $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, $\dim \mathcal{N}(A) = n - r$, $\dim \mathcal{N}(A') = m - r$ (if $r = 0$, then $A = 0$ and $\mathcal{C}(A) = \mathcal{R}(A) = \{0\}$).

Proof.

P and $PA = R$: the row operations multiply on the left by an invertible E (Lecture 2, Theorem 14(c)); column by column (Lecture 1, Theorem 4(a)):

$$E[A \mid I_m] = [EA \mid E] = [R \mid P] \implies P = E \text{ is invertible, } PA = R$$

(a) $\mathcal{C}(A) = \mathrm{span}\{a_1, \ldots, a_n\}$; Theorem 2(b).

(b) $P = (p_{ij})$; Lecture 1, Theorem 4(c); Lecture 3, Theorem 7:

$$\mathrm{row}_i(R) = \mathrm{row}_i(P)A = p_{i1}\,\mathrm{row}_1(A) + \cdots + p_{im}\,\mathrm{row}_m(A) \in \mathcal{R}(A) \implies \mathcal{R}(R) \subseteq \mathcal{R}(A)$$
$$A = P^{-1}R \implies \mathcal{R}(A) \subseteq \mathcal{R}(R) \qquad \text{so } \mathcal{R}(R) = \mathcal{R}(A)$$
$$\text{rows } r+1, \ldots, m \text{ of } R \text{ are } 0 \implies \rho_1', \ldots, \rho_r' \text{ span } \mathcal{R}(R) = \mathcal{R}(A)$$
$$\mathrm{col}_{k_i}(R) = e_i \implies \text{entry } k_i \text{ of } c_1\rho_1' + \cdots + c_r\rho_r' \text{ is } c_i \implies \text{independent}$$

(c) $Ax = 0 \iff Rx = 0$ (Lecture 2, Theorem 3(b)); pivot unknowns from free ones (Lecture 2, Proposition 7(b)); $t \in \mathcal{N}(A)$, $\tilde{t} = t_{f_1}s_1 + \cdots + t_{f_{n-r}}s_{n-r}$:

$$\text{entry } f_l \text{ of } c_1s_1 + \cdots + c_{n-r}s_{n-r} \text{ is } \textstyle\sum_j c_j\delta_{jl} = c_l \implies \text{independent}$$
$$\tilde{t} \in \mathcal{N}(A) \text{ (a subspace)}, \quad \tilde{t}_{f_l} = t_{f_l} \ (l = 1, \ldots, n-r)$$
$$x_{k_i} = -\textstyle\sum_l R_{if_l}x_{f_l} \text{ for every solution} \implies t = \tilde{t} \in \mathrm{span}\{s_1, \ldots, s_{n-r}\}$$
$$r = n: \text{ no free unknowns, so } x = 0 \text{ is the only solution}$$

(d) Lecture 1, Theorems 4(c), 8; Lecture 3, Proposition 12(b):

$$\begin{aligned}
&i > r:\ \operatorname{row}_i(P)A = \operatorname{row}_i(PA) = \operatorname{row}_i(R) = 0 \implies A' \operatorname{row}_i(P)' = (\operatorname{row}_i(P)A)' = 0\\
&c'P = c_1 \operatorname{row}_1(P) + \cdots + c_m \operatorname{row}_m(P) = 0 \implies c' = c'PP^{-1} = 0\\
&\implies \text{the rows of } P \text{ are independent, and so are rows } r+1, \ldots, m\\
&A'y = 0,\ z' = y'P^{-1}: \quad y' = z'P, \quad 0 = y'A = z'PA = z'R = z_1\rho_1' + \cdots + z_r\rho_r'\\
&\implies z_1 = \cdots = z_r = 0 \implies y' = \textstyle\sum_{i>r} z_i \operatorname{row}_i(P) \qquad ((b))
\end{aligned}$$

□

Example 9. Find bases of the four subspaces of $A = \left(\begin{smallmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{smallmatrix}\right)$.

Solution.

Row-reduce $[A \mid I_3]$:

$$\left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 1 & 3 & 0 & 5 & 0 & 1 & 0 \\ 2 & 5 & 1 & 8 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 2 & -2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc|ccc} 1 & 0 & 3 & -1 & 3 & -2 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right) = [R \mid P]$$

$r = 2$, pivot columns $1, 2$, free unknowns x_3, x_4 (already visible in the echelon form after the second arrow; Remark 3).

$\mathcal{C}(A)$, $\mathcal{R}(A)$ (Theorem 8(a),(b)); columns 3, 4 of R (Theorem 2(b)):

$$\begin{aligned}
&\mathcal{C}(A):\ a_1 = (1,1,2)',\ a_2 = (2,3,5)'; \quad a_3 = 3a_1 - a_2, \quad a_4 = -a_1 + 2a_2; \quad \dim \mathcal{C}(A) = 2\\
&\mathcal{R}(A):\ \rho_1' = (1,0,3,-1),\ \rho_2' = (0,1,-1,2); \quad \dim \mathcal{R}(A) = 2
\end{aligned}$$

Check: $3(1,1,2)' - (2,3,5)' = (1,0,1)' = a_3$ and $-(1,1,2)' + 2(2,3,5)' = (3,5,8)' = a_4$.

$\mathcal{N}(A)$, $\mathcal{N}(A')$ (Theorem 8(c),(d)):

$$\begin{aligned}
&Rx = 0:\ x_1 + 3x_3 - x_4 = 0,\ x_2 - x_3 + 2x_4 = 0\\
&\mathcal{N}(A):\ s_1 = (-3,1,1,0)'\ (x_3 = 1,\ x_4 = 0), \quad s_2 = (1,-2,0,1)'\ (x_3 = 0,\ x_4 = 1)\\
&\dim \mathcal{N}(A) = 2 = 4 - 2, \quad \dim \mathcal{N}(A') = 1 = 3 - 2\\
&\mathcal{N}(A'):\ y = \operatorname{row}_3(P)' = (-1,-1,1)'\\
&y'A = -\operatorname{row}_1(A) - \operatorname{row}_2(A) + \operatorname{row}_3(A) = 0: \quad \operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A)
\end{aligned}$$

Warning: every column of R has third entry 0, so every vector of $\mathcal{C}(R)$ does; but a_1 has third entry 2. So $\mathcal{C}(R) \neq \mathcal{C}(A)$: row operations keep $\mathcal{R}(A)$ and $\mathcal{N}(A)$ (Theorem 8(b); Lecture 2, Theorem 3(b)), not $\mathcal{C}(A)$, and a basis of $\mathcal{C}(A)$ consists of pivot columns of A itself, not of R.

3. Rank

Definition 10. By Theorem 8, $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, the number of pivots of an RREF of A (also when $r = 0$): the **rank** $\operatorname{rank} A$ of A; by Remark 3, it is the number of nonzero rows of any echelon form of A. $A \in \mathbb{R}^{m \times n}$ has **full column rank** if $\operatorname{rank} A = n$, **full row rank** if $\operatorname{rank} A = m$. $\dim \mathcal{N}(A)$ is the **nullity** of A.

Theorem 11. For every $A \in \mathbb{R}^{m\times n}$: $\operatorname{rank} A + \dim \mathcal{N}(A) = n$ and $\operatorname{rank} A + \dim \mathcal{N}(A') = m$.

Proof.

R as in Theorem 8, $\operatorname{rank} A = r$:

$$n = r \text{ (pivot columns)} + (n-r) \text{ (free columns)}, \quad \dim \mathcal{N}(A) = n - r \qquad \text{(Theorem 8(c))}$$
$$\dim \mathcal{N}(A') = m - r, \text{ the number of zero rows of } R \qquad \text{(Theorem 8(d))}$$

□

Theorem 12. Let $A \in \mathbb{R}^{m\times n}$.

(a) $\operatorname{rank} A' = \operatorname{rank} A$, and $\operatorname{rank} A \leqslant \min\{m, n\}$.

(b) A has full column rank $\iff$ its columns are linearly independent $\iff$ $\mathcal{N}(A) = \{0\}$.

(c) A has full row rank $\iff$ its rows are linearly independent $\iff$ $\mathcal{N}(A') = \{0\}$ $\iff$ $\mathcal{C}(A) = \mathbb{R}^m$.

(d) If $m = n$: A is invertible $\iff$ $\operatorname{rank} A = n$.

Proof. Let R be an RREF of A, with r nonzero rows and pivot columns $k_1 < \cdots < k_r$; $\operatorname{rank} A = r$ (Definition 10).

(a) $(y'A)' = A'y$ (Lecture 1, Theorem 8): transposing maps $\mathcal{R}(A)$ onto $\mathcal{C}(A')$, keeps linear combinations and is undone by transposing again:

$$\text{a basis of } \mathcal{R}(A) \text{ goes to a basis of } \mathcal{C}(A') \implies \operatorname{rank} A' = \dim \mathcal{C}(A') = \dim \mathcal{R}(A) = \operatorname{rank} A$$
$$r \leqslant m \ (R \text{ has } m \text{ rows}), \quad r \leqslant n \ (k_1 < \cdots < k_r \text{ lie in } \{1, \ldots, n\})$$

(b) Full column rank means $r = n$ (Definition 10); so (b) is Theorem 1: (iii) $\iff$ (i) $\iff$ (ii).

(c) Lecture 1, Theorems 4(c), 8:

$$\operatorname{rank} A = m \iff \dim \mathcal{N}(A') = m - \operatorname{rank} A = 0 \iff \mathcal{N}(A') = \{0\} \qquad \text{(Theorems 11, 4)}$$
$$y'A = y_1 \operatorname{row}_1(A) + \cdots + y_m \operatorname{row}_m(A), \quad y'A = 0 \iff A'y = 0$$
$$\implies \big(\mathcal{N}(A') = \{0\} \iff \text{the rows of } A \text{ are independent}\big)$$
$$\operatorname{rank} A = \dim \mathcal{C}(A) = m \iff \mathcal{C}(A) = \mathbb{R}^m \qquad \text{(Theorems 5(a), 4)}$$

(d) Lecture 2, Theorem 15:

$$A \text{ invertible} \iff (Ax = 0 \implies x = 0) \iff \mathcal{N}(A) = \{0\} \iff \operatorname{rank} A = n \qquad \text{((b))}$$

□

Corollary 13. Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and $[A \mid b] \in \mathbb{R}^{m\times(n+1)}$.

(a) $Ax = b$ is consistent $\iff$ $b \in \mathcal{C}(A)$ $\iff$ $\operatorname{rank}[A \mid b] = \operatorname{rank} A$.

(b) If $Ax = b$ is consistent, its solution is unique $\iff$ $\operatorname{rank} A = n$.

Proof.

(a) $Ax = x_1a_1 + \cdots + x_na_n$ (Lecture 1, Theorem 4(b)):

$$Ax = b \text{ has a solution} \iff b \in \mathcal{C}(A)$$
$$\mathcal{C}(A) \subseteq \mathcal{C}([A \mid b]) \quad \text{(coefficient 0 on } b\text{)}$$
$$b = Ac \implies x_1a_1 + \cdots + x_na_n + tb = A(x + tc) \in \mathcal{C}(A) \implies \mathcal{C}([A \mid b]) = \mathcal{C}(A)$$
$$\operatorname{rank}[A \mid b] = \operatorname{rank} A \implies \mathcal{C}(A) = \mathcal{C}([A \mid b]) \ni b \qquad \text{(Theorem 5(a))}$$

(b) Solutions $x_p + z$, $z \in \mathcal{N}(A)$ (Lecture 2, Theorem 22(a)):

$$\text{unique} \iff \mathcal{N}(A) = \{0\} \iff \operatorname{rank} A = n \qquad \text{(Theorem 12(b))}$$

□

Equations of $\mathcal{C}(A)$. Let row operations take $[A \mid I_m]$ to $[R \mid P]$ as in Theorem 8, $r = \operatorname{rank} A$, and $b \in \mathbb{R}^m$.

The same operations multiply on the left by E (Lecture 2, Theorem 14(c)); Lecture 1, Theorem 4(a); Corollary 13(a); Lecture 2, Proposition 7(a) with $z = Pb$:

$$E[A \mid I_m] = [EA \mid E] = [R \mid P] \implies E = P, \quad E[A \mid b] = [R \mid Pb]$$
$$b \in \mathcal{C}(A) \iff Ax = b \text{ is consistent} \iff \operatorname{row}_i(P)\, b = 0 \quad (i = r+1, \ldots, m)$$

No condition if $r = m$.

Example 14. For A of Example 9, when is $Ax = (1, t, 4)'$ consistent? Find all its solutions then.

Solution.

Example 9: $r = 2$, $\operatorname{row}_3(P) = (-1, -1, 1)$; the equations of $\mathcal{C}(A)$:

$$b \in \mathcal{C}(A) \iff -b_1 - b_2 + b_3 = 0 \quad (\text{the coefficients of } \operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A))$$
$$b = (1, t, 4)' : \ -1 - t + 4 = 0 \iff t = 3$$

$t = 3$, $b = (1, 3, 4)'$; $Ax = b \iff Rx = Pb$ (Lecture 2, Theorem 3(b)); all solutions by Lecture 2, Theorem 22(a), and Theorem 8(c):

$$Pb = (3 - 6, \ -1 + 3, \ -1 - 3 + 4)' = (-3, 2, 0)'$$
$$Rx = Pb : \ x_1 + 3x_3 - x_4 = -3, \ x_2 - x_3 + 2x_4 = 2; \quad x_3 = x_4 = 0 : \ x_p = (-3, 2, 0, 0)'$$
$$\text{all solutions: } x_p + c_1 s_1 + c_2 s_2 \ (c_1, c_2 \in \mathbb{R}); \quad \text{not unique: } \operatorname{rank} A = 2 < 4 \text{ (Corollary 13(b))}$$

Check: $Ax_p = -3a_1 + 2a_2 = (-3 + 4, \ -3 + 6, \ -6 + 10)' = (1, 3, 4)'$.

4. Rank equalities and inequalities

Theorem 15. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$.

(a) $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$, $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$, and $\operatorname{rank}(AB) \leqslant \min\{\operatorname{rank} A, \operatorname{rank} B\}$.

(b) If A has full column rank, then $\mathcal{N}(AB) = \mathcal{N}(B)$ and $\operatorname{rank}(AB) = \operatorname{rank} B$.

(c) If B has full row rank, then $\mathcal{C}(AB) = \mathcal{C}(A)$ and $\operatorname{rank}(AB) = \operatorname{rank} A$.

(d) $\operatorname{rank}(PAQ) = \operatorname{rank} A$ for all invertible $P \in \mathbb{R}^{m \times m}$ and $Q \in \mathbb{R}^{n \times n}$.

Proof.

(a) $(AB)' = B'A'$ (Lecture 1, Theorem 8); Theorem 12(a):

$$ABx = A(Bx) \in \mathcal{C}(A); \quad Bx = 0 \implies ABx = A0 = 0$$
$$\operatorname{rank}(AB) = \dim \mathcal{C}(AB) \leqslant \dim \mathcal{C}(A) = \operatorname{rank} A \qquad \text{(Theorem 5(a))}$$
$$\operatorname{rank}(AB) = \operatorname{rank}\big((AB)'\big) = \operatorname{rank}(B'A') \leqslant \operatorname{rank} B' = \operatorname{rank} B$$

(b) $\mathcal{N}(A) = \{0\}$ (Theorem 12(b)):

$$ABx = 0 \implies Bx \in \mathcal{N}(A) = \{0\} \implies x \in \mathcal{N}(B); \quad \text{with (a), } \mathcal{N}(AB) = \mathcal{N}(B)$$
$$\operatorname{rank}(AB) = p - \dim \mathcal{N}(AB) = p - \dim \mathcal{N}(B) = \operatorname{rank} B \qquad \text{(Theorem 11)}$$

(c) $\mathcal{C}(B) = \mathbb{R}^n$ (Theorem 12(c)):

$$x \in \mathbb{R}^n \implies x = Bw \implies Ax = (AB)w \in \mathcal{C}(AB) \implies \mathcal{C}(A) \subseteq \mathcal{C}(AB)$$
$$\text{with (a), } \mathcal{C}(AB) = \mathcal{C}(A), \text{ so } \operatorname{rank}(AB) = \operatorname{rank} A$$

(d) Theorem 12(b),(d):

$$Px = 0 \implies x = P^{-1}Px = 0 : \ P \text{ has full column rank}$$
$$\operatorname{rank} Q = n : \ Q \text{ has full row rank}$$
$$\operatorname{rank}(P(AQ)) = \operatorname{rank}(AQ) = \operatorname{rank} A \qquad ((b), (c))$$

□

Theorem 16. Let $A \in \mathbb{R}^{m\times n}$.

(a) $\mathcal{N}(A'A) = \mathcal{N}(A)$.

(b) $\operatorname{rank}(A'A) = \operatorname{rank} A = \operatorname{rank}(AA')$.

(c) $\mathcal{C}(A'A) = \mathcal{C}(A')$ and $\mathcal{C}(AA') = \mathcal{C}(A)$.

Proof.

(a) Lecture 1, Theorem 8:

$Ax = 0 \implies A'Ax = A'0 = 0$

$A'Ax = 0 \implies (Ax)'(Ax) = x'A'Ax = 0,$ the sum of the squares of the entries of $Ax \implies Ax = 0$

(b) $A'A$ and A have n columns; Theorem 11:

$$\operatorname{rank}(A'A) = n - \dim \mathcal{N}(A'A) = n - \dim \mathcal{N}(A) = \operatorname{rank} A \qquad ((a))$$
$$A' \text{ in place of } A : \ \operatorname{rank}(AA') = \operatorname{rank} A' = \operatorname{rank} A \qquad ((A')' = A;\ \text{Theorem 12(a)})$$

(c) Theorems 15(a), 5(a):

$$\mathcal{C}(A'A) \subseteq \mathcal{C}(A'), \text{ both of dimension } \operatorname{rank} A \ ((b), \text{Theorem 12(a)}) \implies \mathcal{C}(A'A) = \mathcal{C}(A')$$
$$\mathcal{C}(AA') \subseteq \mathcal{C}(A), \text{ both of dimension } \operatorname{rank} A \ ((b)) \implies \mathcal{C}(AA') = \mathcal{C}(A)$$

□

So $A'A \in \mathbb{R}^{n\times n}$ is invertible $\iff \operatorname{rank} A = n$ (Theorems 16(b), 12(d)).

Example 17. $A = \operatorname{diag}(1,1,1,0)$ and $B = \operatorname{diag}(0,0,1,1)$ in $\mathbb{R}^{4\times 4}$: find $\operatorname{rank} A$, $\operatorname{rank} B$, $\operatorname{rank}(AB)$ and $\operatorname{rank}(A'A)$.

Solution.

A zero column adds nothing to the span (Lecture 3, Theorem 9(c)); unit vectors are independent (Lecture 3, Definition 14, Proposition 12(b)):

$$A = [e_1\ e_2\ e_3\ 0] : \ \mathcal{C}(A) = \operatorname{span}\{e_1, e_2, e_3\}, \ \operatorname{rank} A = 3$$
$$B = [0\ 0\ e_3\ e_4] : \ \mathcal{C}(B) = \operatorname{span}\{e_3, e_4\}, \ \operatorname{rank} B = 2$$
$$\operatorname{row}_i(AB) = a_{ii} \operatorname{row}_i(B) \qquad \text{(Lecture 1, Theorem 4(c))}$$
$$AB = \operatorname{diag}(0,0,1,0) = [0\ 0\ e_3\ 0] : \ \operatorname{rank}(AB) = 1 < \min\{3,2\} \qquad \text{(Theorem 15(a))}$$
$$A' = A \text{ (diagonal)}, \ A^2 = A \ (\operatorname{row}_i(A^2) = a_{ii} \operatorname{row}_i(A)) \qquad \text{(Lecture 1, Definition 7)}$$
$$A'A = A^2 = A : \ \operatorname{rank}(A'A) = 3 = \operatorname{rank} A \qquad \text{(Theorem 16(b))}$$

Practice. The Exercises and Problems of the textbook, Chapter 4: bases and dimension, the four subspaces, rank, consistency of $Ax = b$, and the rank of products and of $A'A$.